

1. Transitioned to an EPA-Driven Architecture

Overview

Fundamentally, the machine learning pipeline is re-architected to optimize for Expected Points Added (EPA) rather than raw yards gained. Yards can be highly deceptive (e.g., a 10-yard completion on 3rd & 15 results in a punt, which is mathematically negative for scoring probability). By targeting EPA, the AI now evaluates the true scoring probability of every play call based on down, distance, and field position.

Code Changes

- **Backend (Training Scripts):** In `exp_run_yards_model.py` and `exp_pass_yards_model.py`, the target variable was swapped from `y = df['yards_gained']` to `y = df['epa']`.
- **Frontend (React):** In `result.js`, the data labels are updated to accurately reflect the new math, changing the hardcoded `<h2>Expected Yards: ...</h2>` to `<h2>Expected Points Added (EPA): ...</h2>` so the UI syncs with the backend calculations.

Future Roadmap

The plan is to introduce EPA "clipping" (capping the max EPA reward at +2.0) during model training to prevent the AI from over-indexing on deep, explosive routes (like Y-Cross) at the expense of consistent, chain-moving concepts. We can also aim to integrate Win Probability Added (WPA) as a secondary evaluation metric for late-game, clock-management scenarios.